

Title

The impact of deindustrialization on the proliferation of abandoned structures in
Pennsylvania

Research Question

What is the role of deindustrialization on the increase of abandoned structures in
Pennsylvania?

Word Count

3491

Table of Contents

Introduction	2
The Effect on the Steel Industry	3
The Environmental Impact of Bethlehem Steel	5
The Effect on the Automotive Industry	6
The Effect on Residential Housing & Population Size	8
Case Studies of Residential Decline: Brownsville, Clairton	9
The Effect on the Environment	11
Case Studies of Environmental Decline: Philadelphia Generating Stations	12
Conclusion	14
Bibliography	16

Introduction

This Extended Essay will answer the question, **What is the role of deindustrialization on the increase of abandoned structures in Pennsylvania?** To answer this question, a comprehensive overview of factors leading to deindustrialization must be understood. In the mid-1960s, the United States was in the midst of a complex and multi-faceted economic situation characterized by newfound prosperity and quickly emerging challenges. The nation experienced remarkable post-war growth in many aspects. Economic activity flourished as technological advancements, increased consumer spending, and a wealthier middle class emerged. As a result of this increased consumerism, the American economy continued to emphasize mass production as they did in the world wars. There was a need to meet the rising demands of a thriving consumer culture. From the outside, America was doing better than ever, with a well-off middle class and increased ownership of ordinary pleasures like automobiles. The market appeared to be in a place of power and sustenance. However, there were indications of shifting market dynamics beneath the surface of prosperity. Pressure on domestic industry grew as the global economic landscape did. There was a need for strategic adaptation and change that the US should have pursued. The same factors that contributed to initial financial success in mass production and focus on industry posed many challenges. The struggles of the domestic economy led to a shift in consumer behavior and a change to the global economic structure. This shift initiated a period of deindustrialization, a systematic decline, and a reduction in the contribution of manufacturing and industrial pursuits to the economy. Deindustrialization has had an immeasurable toll on the communities, businesses, residences, and environmental quality of the United States (U.S.). Around 1965, the U.S. began a switch to a global market, as rates of domestic interest and exporting costs skyrocketed. As a result, it became significantly more expensive to buy U.S.-manufactured products than imported products from other countries. Both foreigners

and natives stopped supporting the U.S. manufacturing and industrial industries, leading to a decrease in the demand for domestic production.

Automotive, steel, coal, and manufacturing industries provided materials that were no longer necessary to the same extent. The area that most absorbed the impact of this decline was the Northeastern United States, stretching horizontally from Illinois to Pennsylvania and vertically from Tennessee to New York. This region was the country's manufacturing, steel-making, and coal-producing heartland, known as the Rust Belt. As devastating unemployment, growing poverty, and shrinking tax bases shut down the national economy, the doors to the Rust Belt's mills, factories, mines, plants, and industrial facilities abruptly closed. There was no longer a demand for the extensive resources and employees that the nation had once required. **Migration to a global economy and subsequent deindustrialization of manufacturing-dependent eastern states induced a drastic increase in nearby abandoned structures, many of which still litter the region - decaying, rusting, and crumbling, with no plan for restoration.** Due to this deindustrialization of the nation, Pennsylvania, a former hub of industry, is flooded with an excess of abandoned residences, industrial facilities, manufacturing buildings, and unusable land.

The Effect on the Steel Industry

The decline in demand for steel following World War II led to the downfall of the domestic steel industry, which, in turn, initiated a widespread economic downturn and spurred the proliferation of various abandoned structures in the Rust Belt region. In 1955, the United States supplied roughly 40 percent of the world's steel needs and imported only 1 percent of its domestic consumption. The steel industry provided close to 1 percent of total civilian employment in the United States at that time. By 1985, however, the US steel

industry accounted for only 11 percent of the world's steel; more than 25 percent of US consumption was imported; and the US steel industry provided less than 0.2 percent of total civilian employment.¹ This downward trend can be partially attributed to the nation's history as the primary steel provider in the world. During both World Wars, the US was focused on mass production rather than research and innovation. Other nations concentrated on technological advancements in their steel production processes, thus establishing more developed and economically efficient production methods post-war. By 1972, the competitive position of US producers had deteriorated to the point where domestic unit operating costs were about 40 percent higher than those of Japanese producers.² This decline was most endured in the Rust Belt, a former steel production hub.

One of the most prolific steel producers of the twentieth century was Bethlehem Steel, headquartered in Bethlehem, Pennsylvania. Established in 1899, Bethlehem Steel was a quickly growing company that was pivotal in the foundation of American steel. During WWI and WW2, the company was the primary provider of armor plates and weaponry, frequently called the "most important to America's national defense of any company in the past century."³ Bethlehem Steel also played a vital role in the U.S.'s shipbuilding industry, producing around ⅓ of the Navy's fleet, more than any other provider in the nation.⁴ However, as the steel industry declined, so did Bethlehem Steel. In 1982, Bethlehem Steel reported a loss of US\$1.5 billion and shut down much of its operations. The main steel production plant faced full closure a decade later.⁵ The quick fall of Bethlehem Steel illustrates the fast decline

¹ IMF. External Relations Dept, "Finance & Development, December 1987," *Finance & Development* 24, no. 4 (1987): [Pages 1-52], <https://doi.org/10.5089/9781616353711.022>.

² IMF. External Relations Dept, "Finance & Development," [Page #].

³ Kirstin Downey, "Bethlehem Steel Corp. Is No More," *The Washington Post*, last modified May 1, 2003, accessed November 3, 2023, <https://www.washingtonpost.com/archive/business/2003/05/01/bethlehem-steel-corp-is-no-more/55b6daa3-9016-4ebd-9b27-b4b8c704afbc/>.

⁴ Nathan Brodsky, "The Weapons Acquisition Process — an Economic Analysis, by Merton J. Peck and Frederic M. Scherer. Boston: Division of Research, Harvard Business School, 1962. Pp. 736," *Naval Research Logistics Quarterly* 9, nos. 3-4 (1962): [Page #], <https://doi.org/10.1002/nav.3800090313>.

⁵ "Bethlehem Steel, The People Who Built America," video, 1:19:27, YouTube, posted by PBS39, February 5, 2008, accessed January 29, 2024, <https://www.youtube.com/watch?v=2QTGiHOZZFU>.

of the US steel industry as deindustrialization began. This primary provider of what was one of America's most valuable resources closed just 20 years after it was valued as the most profitable business in the nation.

The Environmental Impact of Bethlehem Steel

Bethlehem Steel's extensive steel production led to excess industrial waste deposited in nearby water sources and land. The area surrounding the plant consists of approximately 489 acres where iron, steel-making slag, and other manufacturing wastes from the production plant were deposited.⁶ Much of the 1000-acre site has been affected by this waste, resulting in a deed restriction banning residential use due to extensive contamination. The deed restriction limits much of the potential for reuse and renovation of the site.⁷ Additionally, to clean the land only to the extent it would be suitable for areas of less prolonged exposure, such as new manufacturing facilities or recreational fields, would cost an estimated 200 million USD.⁸ The demolition and removal of industrial equipment also presents a high cost. Components of Bethlehem Steel are reported to contain contaminants that are costly to remove such as asbestos, lead, and arsenic.⁹ The sheer size and density of Bethlehem Steel would most definitely ensure an expensive and time-consuming effort for reuse. These factors illustrate the reason for the continued abandonment of Bethlehem Steel and other industrial facilities forced into closure by deindustrialization. The steel-making facilities are no longer necessary

⁶ Parsons, "Frequently Asked Questions," Former Bethlehem Steel Site Cleanup, accessed January 29, 2024, <https://bethlehemsteelcleanup.com/faq/>

⁷ CITY OF BETHLEHEM, "Comprehensive Plan 2008 Future Land Use," Bethlehem Pennsylvania, last modified 2008, accessed January 29, 2024, https://www.bethlehem-pa.gov/CityOfBethlehem/media/DCED-Media-Library/PlanningZoning/CommunityPlans/Comprehensive_Plan_2009_Future_Land_Use.pdf

⁸ Mike Desmond, "Agreement expected this summer for \$69-\$200M Bethlehem Steel cleanup," WBFO, last modified May 19, 2021, accessed January 29, 2024, <https://www.wbfo.org/environment/2021-05-19/agreement-expected-this-summer-for-69-200m-bethlehem-steel-cleanup>

⁹ Aaron Munz, "Bethlehem Steel Shipyard," ed. Walter Pacheco, Asbestos.com, last modified August 25, 2023, accessed January 29, 2024, <https://www.asbestos.com/shipyards/bethlehem-steel/>

and incredibly expensive to convert into other zoned properties. The combination of environmental damages at these sites makes repurposing cost-ineffective and undesirable.

The Effect on the Automotive Industry

The decline in manufacturing substantially affected the automotive industry and subsequently increased the amount of abandoned auto-production facilities in the Rust Belt. For decades, the US had led in global automotive production, producing more than 90% of total automobiles in 1929. However, this began to decline quickly. By 1945, the US produced roughly 75% of automobiles, and in 1980 they produced around 30%.¹⁰ This sharp decrease can be attributed to the growing import automaker share. Previously, American car manufacturers such as Chrysler, General Motors, and Ford were primary providers.¹¹ These principal companies were headquartered around the Rust Belt, relying on natural sources of iron and aluminum. German automobile manufacturers such as Volkswagen, Porsche, and BMW also focused efforts on manufacturing cars within the Northeast US to take advantage of natural resources and market availability.¹² Contrarily, Japanese companies focused on manufacturing domestically and then importing vehicles to the US. Japan systematically approached car manufacturing, focusing on affordability and reliability over appearance. Like with the steel industry, Japan focused on optimization and modernization rather than mass production. As a result, they overtook America as the primary car manufacturer in 1980, which they maintained for over a decade.¹³

¹⁰ November 1929, Popular Science, 126, accessed January 29, 2024, <https://books.google.com/books?id=FSgDAAAAMBAJ&q>.

¹¹ Dee-Ann Durbin, "Toyota passes Ford as 2nd-biggest auto seller in US in 2007," LA Times, last modified January 3, 2008, accessed January 29, 2024, <https://www.latimes.com/business/autos/la-ap-mi-autosales-story.html>.

¹² Stephen J. Silvia, "ORGANIZING GERMAN AUTOMOBILE PLANTS IN THE USA - An Assessment of the United Auto Workers' Efforts to Organize German-Owned Automobile Plants," Hans Bockler Stiftung, last modified December 2016, accessed January 29, 2024, https://www.boeckler.de/fpdf/HBS-006530/p_study_hbs_349.pdf

¹³ Michael A. Cusumano, "Manufacturing Innovation: Lessons from the Japanese Auto Industry," MIT Sloan Management Review, last modified October 15, 1988, accessed January 29, 2024,

The decline of the American automotive industry resulted in the abandonment and financial downfall of many US automotive production facilities. A prime example is the Volkswagen Westmoreland Assembly Plant near Pittsburgh, Pennsylvania. Operating until 1987, Westmoreland Assembly produced over 1 million Volkswagens.¹⁴ The plant was introduced near Pittsburgh to capitalize upon nearby sources of essential materials and manufacturing resources. As these materials became more costly to acquire domestically, Westmoreland Assembly faced troubles. Much like American automakers of the time, the company struggled to compete with Japanese manufacturer prices. The Volkswagen Rabbit, the vehicle primarily produced in Westmoreland, cost \$2000 more than its Japanese counterpart, the Honda Civic.¹⁵ Due to pressures from the Japanese auto industry and the increasing costs of US manufacturing, Volkswagen moved production out of the US. Westmoreland Assembly aligned with the company's troubles, as they noted a roughly 120 million dollar annual loss before closure. The plant was officially closed and abandoned in 1988, marking a closure of all non-Japanese foreign automakers in the US.¹⁶ Westmoreland Assembly was one of an estimated 267 automotive facilities within the Rust Belt that closed during the period of deindustrialization in the United States. The sudden decline of the demand for domestic automotive production resulted in these structures consuming areas near Pittsburgh and central Pennsylvania. Less than 40% of these abandoned sites have been repurposed since their closure, with less than ten being repurposed for recreational, governmental, or residential usage.¹⁷

<https://sloanreview.mit.edu/article/manufacturing-innovation-lessons-from-the-japanese-auto-industry/>.

¹⁴ Tim Moran, "VW production tripped up in US - Many problems lead to automaker's closing of PA factory," *Automotive News*, last modified October 17, 2005, accessed January 29, 2024, <https://www.autonews.com/article/20051017/SUB/510170703/vw-production-tripped-up-in-u-s>.

¹⁵ Moran, "VW production," *Automotive News*.

¹⁶ William Beaver, "Volkswagen's American Assembly plant: fahrvergnugen was not enough," *Business Horizons*, last modified December 1992, accessed January 29, 2024.

¹⁷ Sean Culey, "Revitalizing the Rust Belt," *Forbes*, last modified September 8, 2017, accessed January 29, 2024, <https://www.forbes.com/sites/realspin/2017/09/08/revitalizing-the-rust-belt/>.

The Effect on Residential Housing & Population Size

The shift from a domestic to a global market and subsequent abandonment of industrial and manufacturing facilities led to the abandonment of nearby residential housing. As the industry shifted, employment opportunities sharply declined in the Rust Belt. Primary job providers in steel, automotive, and other production industries shut down. The towns that relied on these industries faced mass unemployment, and residents fled from the desperate and undesirable areas. The decline in the need for labor forced workers out of these towns. Primary cities for steel and manufacturing industries reported population declines of at least 20% per decade between 1960 and 1990.¹⁸ The uprooting of so many individuals caused housing to be left vacant and unwanted across the Rust Belt. These populations migrated to the Sun Belt, an area stretching from the Southwest to Southeast US.¹⁹ These regions provided growing economic opportunities with an influx of modern technological advancements. The oil, defense, and aerospace industries capitalized on innovation in the South and offered extensive job opportunities. Thus, the Rust Belt was left undesirable and in ruin.

Deindustrialization has led to abandoned structures in Pittsburgh that still impact the city today. Pittsburgh was one of the primary producing steel cities in the nation during the industrial boom, with steel mills running 24 hours a day and requiring massive workforces to sustain them. When the need for steel declined, so did the demand for labor. The city's population declined by over 50% between 1940 and 1990, leading to a need for only 50% of

¹⁸ Alan Berube and Cecile Murray, "Renewing America's economic promise through OLDER INDUSTRIAL CITIES," Metropolitan Policy Program at Brookings, last modified April 2018, accessed January 29, 2024, https://www.brookings.edu/wp-content/uploads/2018/04/2018-04_brookings-metro_older-industrial-cities_full-report-berube_murray_final-version_af4-18.pdf.

¹⁹ "Sunbelt Migration," University of Central Florida, accessed January 29, 2024, <https://vlp.cah.ucf.edu/popups/sunbelt.html>.

previously occupied housing facilities to be maintained.²⁰ This area was in economic despair and did not attract new residents or industries to fill the place of vacant structures. Without the population they had sustained, the city also lost extensive commercial, recreational, and governmental spaces. Certain communities in the city have lost over 75% of their populations since the decline. In 2022, Pittsburgh was reported to have over 50,000 vacant properties, making up around 10% of available units.²¹ These vacant properties lead to over 10 million dollars in police and fire fees and 8 million in delinquent property taxes. The unattractive structures have also been reportedly stolen nearby property values of over 250 million dollars annually.²²

Case Studies of Residential Decline: Brownsville, Clairton

Two of the Rust Belt's most famous "ghost towns" lie on the outskirts of Pittsburgh, left in ruin due to rapid deindustrialization. Brownsville, Pennsylvania, is one of these frighteningly empty towns dotting the region. Brownsville is a section of Pittsburgh located on the Ohio River. Convenient access to the river led to the rapid growth of industry, commerce, and transportation within the town. In fact, between 1930 and 1940, Brownsville incurred a population increase of 179%, growing from roughly 2,000 to 9,000.²³ Brownsville's steel and coal industries provided countless job opportunities for residents, stimulating nearby commercial and industrial growth and attracting residents. But in the

²⁰ "The Root of Pittsburgh's Population Drain," Carnegie Mellon Heinz School Center for Economic Development, last modified November 2003, accessed January 29, 2024, <https://www.heinz.cmu.edu/ced/file/pop-drain.pdf>

²¹ Department of City Planning, "Pathways to Removing Obstacles to Housing (PRO Housing)," The City of Pittsburgh, accessed January 29, 2024, <https://engage.pittsburghpa.gov/pathways-removing-obstacles-housing>

²² Center for Community Progress Report to the Urban Redevelopment Authority of Pittsburgh, Pennsylvania, "THE COST OF VACANT AND BLIGHTED PROPERTIES IN PITTSBURGH: A Conservative Analysis of Service, Tax Delinquency, and Spillover Costs," Center for Community Progress, last modified March 2017, accessed January 29, 2024, <https://pghlandbank.org/wp-content/uploads/2022/03/Cost-of-Blight-2.pdf>

²³ U.S. Census Bureau, "Decennial Census of Population and Housing," U.S. Census Bureau, last modified December 11, 2013, <https://www.census.gov/programs-surveys/decennial-census.html>.

1970s, the slump in the steel and coal industries erased jobs in Brownsville. The population steadily decreased by 20% each decade after, leaving countless abandoned residences and commercial facilities.²⁴ Once a booming town, the abandonment of Brownsville left three massive national banks, a skyscraping flatiron building, numerous malls, and many other structures vacant. One of Pennsylvania's most notorious abandoned buildings, the Brownsville General Hospital, remains a sight to see. The hospital had over a 200 bed capacity, and three grandly designed floors but is now left crumbling in the center of the town.²⁵

Clairton, Pennsylvania, is another ghost town on the outskirts of the Ohio River, left vacant from the rapid deindustrialization of the nation. Clairton was the home of the Crucible Steel Company. Crucible Steel was a merger of 13 top-producing steel companies in the US. Crucible was the largest producer of many varieties of steel.²⁶ For a period of time, they were also owned by Colt Industries, a primary weapon producer during both World Wars. But in the 1900s, Crucible Steel closed over 400 US plants and subsequently laid off more than 200,000 workers.²⁷ These closures included the corporation's primary plant in Clairton, Pennsylvania. A plant of US Steel, the largest steel manufacturer in the nation, also resided in Clairton. The closure of these primary employment providers incited a mass migration from Clairton beginning in the 1970s. The population dropped from a peak of 20,000 in 1950 to less than 9,000 in the 1990s.²⁸ These declines spurred a significant economic shock on the community of Clairton. The town is still listed as a distressed municipality by the Department

²⁴ U.S. Census Bureau, "Decennial Census,"

²⁵ "Brownsville General Hospital," Abandoned Online, accessed January 29, 2024, <https://abandonedonline.net/location/brownsville-general-hospital/>

²⁶ The Post Standard. "A timeline of Crucible Materials Corp." Syracuse.com. Last modified May 6, 2009. Accessed January 29, 2024. https://blog.syracuse.com/indepth/2009/05/a_timeline_of_crucible_materia.html

²⁷ Rachel Donaldson, Ph.D, "Labor History in the United States," National Parks Service U.S. Department of the Interior, last modified January 2022, accessed January 29, 2024, https://portal.ct.gov/-/media/DECD/Historic-Preservation/03_Technical_Assistance_Research/Research/Labor_History_in_US-Theme_Study-Final-2.pdf

²⁸ U.S. Census Bureau, "Decennial Census,"

of Community Affairs. As of the 2000 census, over 20% of the population was below the poverty line, and the median household income was under \$25,000.²⁹ The town is home to a notoriously eerie neighborhood known as Lincoln Way that remains entirely abandoned, with over 50 vacant homes on the street.³⁰

The Effect on the Environment

Much like the once-flourishing communities of the nation, the environment was also not protected during the nations' rapid industrialization. Growing industry and commercialism in the Rust Belt occurred before there was time to place environmental legislations or restrictions. Minimal environmental standards were enacted upon industry until the 1980s. The absence of guidelines meant that industries were not compelled to follow emissions, waste disposal, or pollution control standards. Companies often release pollutants directly into the air and nearby water bodies, contaminating surrounding areas. By the end of the 1960s, more than 40% of public water systems couldn't meet minimal health standards.³¹ The lack of guidelines allowed for unrestricted industrial activity that more often than not, resulted in irreversible environmental damage. For example, the Superfund program, focused on addressing contaminated sites nationwide, was not introduced until 1980 as part of a broader initiative, the Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA).³² Before the introduction of this critical legislation, there were no

²⁹ World Population Review, <https://worldpopulationreview.com/us-cities/clairton-pa-population>.

³⁰ Matthew Christopher, "Lincoln Way," Abandoned America, last modified May 15, 2023, accessed January 29, 2024, <https://www.abandonedamerica.us/lincoln-way>.

³¹ Richard Weinmeyer, JD, MA, Ph.D, "The Safe Drinking Water Act of 1974 and Its Role in Providing Access to Safe Drinking Water in the United States," AMA Journal of Ethics, last modified October 2017, accessed January 29, 2024, <https://journalofethics.ama-assn.org/article/safe-drinking-water-act-1974-and-its-role-providing-access-safe-drinking-water-united-states/2017-10>

³² US Government, "Summary of the Comprehensive Environmental Response, Compensation, and Liability Act (Superfund)," Environmental Protection Agency, last modified September 6, 2023, accessed January 29, 2024, <https://www.epa.gov/laws-regulations/summary-comprehensive-environmental-response-compensation-and-liability-act>.

comprehensive guidelines for cleaning and treating industrial sites. In addition to the lack of waste disposal and cleanup legislation, there was a lack of land use planning, as industries established facilities without considering the long-term effects of their actions or consequences to surrounding communities.³³ The quick deindustrialization of the nation then left countless superfund sites and brownfields around previously booming areas. With no motivation to reoccupy or repurpose these non-necessary sites, minimal sanitation occurred. Areas of land were left ravaged by their industrial past with highly contaminated and worthless land.

Case Studies of Environmental Decline: Philadelphia Generating Stations

Philadelphia, Pennsylvania, is home to many of the Rust Belt's vacant superfund and brownfield sites. Both during and after the Industrial Revolution, the city of Brotherly Love was a key provider in the manufacturing and power-generating industries.³⁴ Anyone who takes a trip along the notoriously disgusting Delaware River, will surely notice an immeasurable amount of vacant buildings towering over them. One of which is the Delaware River Generating Station. After WW1, Philadelphia needed more power than it had previously produced.³⁵ So, they hired a famous architect to craft them a fully concrete, large-scale power plant, the first of its kind on the planet. The plant was completed in 1923 and quickly rose to its demands. By the end of its first operating year, Delaware River Generating Station was producing over 180,000 kw, which provided over half of the city's power supply.³⁶ However, the city was still in need of more power. So, plans quickly started

³³ US Government, "Summary of the Comprehensive," Environmental Protection Agency.

³⁴ "Philadelphia in the late 1800s: International Industrial Leader – Then and Now?," Historical Society of Pennsylvania, last modified November 22, 2017, accessed January 29, 2024, <https://hsp.org/blogs/educators-blog/philadelphia-late-1800s-international-industrial-leader-%E2%80%93-then-and-now>.

³⁵ Walter Licht, "Workshop of the World," The Encyclopedia of Greater Philadelphia, accessed January 29, 2024, <https://philadelphiaencyclopedia.org/themes/workshop-of-the-world/>.

³⁶ Matthew Christopher, "Delaware River Generating Station," Abandoned America, last modified September 24, 2021, accessed January 29, 2024, <https://www.abandonedamerica.us/delaware-river-generating-station>.

to build the nearby Richmond Generating Station, just 3 miles down the river. The new plant opened two years after its predecessor, boasting the largest turbines in generating station history.³⁷ These plants were celebrated by citizens, politicians, and industry alike. Described as some of the most historically boastful and impressive feats of industry, it was surprising that they would meet their final days by the 1980s.³⁸ These generating stations lay waste along the rivers of Philadelphia. Countless plans for repurposing have been posed, but none have succeeded.³⁹

The impressive feats of engineering that are the power plants of the Delaware River were laid to waste as the population of Philadelphia rapidly declined and those who remained switched to newer, cleaner sources of energy. Between 1950 and 1980, the city lost nearly 30% of its population.⁴⁰ Former occupants fled to areas of promise and innovation. The Remaining occupants started relying on new forms of energy, such as nuclear, which the nearby Limerick Nuclear Power Plant pioneered. Much like many buildings vacant around the Rust Belt, the impressive generating stations were now obsolete. However, their extensive and practically irreversible contamination will never be forgotten. The abundance of power plants on the Delaware River has been traced to numerous pollution-related damages, such as increased congenital disabilities, marine animal deaths, and declining water quality. A singular generating station on the river is estimated to kill over 14 billion fish, eggs, and larvae each year, decimating species diversity and population.⁴¹ Environmental officials representing a region directly downwind of these generating stations have been in a legal

³⁷ Chris Mele, "Power plants on the Delaware: Abandoned but still celebrated," Delaware Currents, last modified November 22, 2021, accessed January 29, 2024, <https://delawarecurrents.org/2021/11/22/power-plants-on-the-delaware-abandoned-but-still-celebrated>

³⁸ Mele, "Power plants," Delaware Currents.

³⁹ Ashley Hahn, "New chapter possible for Delaware Generating Station," WHYY PBS, last modified January 20, 2015, accessed January 29, 2024, <https://whyy.org/articles/new-chapter-possible-for-delaware-generating-station/>.

⁴⁰ U.S. Census Bureau, "Decennial Census,"

⁴¹ RMC-Environmental Services, "Limerick Generating Station Units 1 & 2," License Renewal Company, last modified December 1985, accessed January 29, 2024, <https://www.nrc.gov/docs/ML1211/ML12110A289.pdf>.

battle with the state for over a decade. They cite a significant increase in children being born at extremely low birth weights, claiming nearby air and water pollution as the source. Officials also report incredibly high asthma and lung-issue rates.⁴² The area continues to fail clean air standards even decades after the closing of these stations. The contamination of this once-thriving coal-dominated city has led to decreased property values for hundreds of miles around and the continued vacancy of many structures. Abandoned buildings can be found littering the city, with an estimated 50,000 vacant structures remaining.⁴³ The effects of these generating stations on nearby communities can be directly attributed to a lack of environmental legislation during the period of American industrialization. Unclear and unestablished guidelines for environmentally responsible practice needed to be improved.

Conclusion

The impact of deindustrialization on abandoned structures in Pennsylvania is a complex issue. The decline of critical industries such as steel and automotive manufacturing triggered an irreversible cascade of economic and environmental struggles. The once vibrant industrial heartland of the nation, exemplified by iconic structures such as Bethlehem Steel and the Delaware and Richmond Generating Stations, succumbed to the forces of a shifting global market and rapidly developing industry dynamics. The repercussions of these shifts were not confined to the economic sector. They reverberated through communities, leading to mass unemployment, population decline, and the subsequent proliferation of abandoned residences and once-enjoyed commercial spaces. Buildings that evoke notions of happiness

⁴² Ioannis Manisalidis et al., "Environmental and Health Impacts of Air Pollution: A Review," *Frontiers in Public Health* 8 (February 20, 2020): <https://doi.org/10.3389/fpubh.2020.00014>.

⁴³ Kristen Monsbrucker-Garza, "Few are working full-time in the office anymore, but some argue it's nowhere near a 'real estate recession' in Philly," WHYY PBS, last modified January 3, 2024, accessed January 29, 2024, <https://whyy.org/articles/center-city-office-market-real-estate-recession-2024-forecast/>

and life, like local shopping malls and elementary schools, remain decaying, crumbling, with no life in them at all.

The extensive environmental toll of the region's rapid industrialization, in conjunction with a widespread absence of early regulatory legislation, left lasting scars on the Rust Belt's landscape. The remnants of this era, focused on mass production and prosperity without considering the environment, still plague the country. Superfund sites and brownfields litter the once-booming areas, further complicating repurposing and reinvigorating these regions. Contaminated properties such as the Delaware River Generating Station shed light on the challenges posed by these superfund sites, emphasizing the extensive impact on public health and property values.

The nation has been ravaged by its own decisions: the decision to proceed without care for its environment, the decision to leave historically significant sites crumbling with no plans for repair, and the ignorance that led to the devastation of industry-reliant towns and their desperate citizens. A conscious, collaborative, and comprehensive decision must be made to address the legacy of deindustrialization.

Bibliography

- Beaver, William. "Volkswagen's American Assembly plant: fahrvergnugen was not enough." Business Horizons. Last modified December 1992. Accessed January 29, 2024.
- Berube, Alan, and Cecile Murray. "Renewing America's economic promise through OLDER INDUSTRIAL CITIES." Metropolitan Policy Program at Brookings. Last modified April 2018. Accessed January 29, 2024.
https://www.brookings.edu/wp-content/uploads/2018/04/2018-04_brookings-metro_older-industrial-cities_full-report-berube_murray_final-version_af4-18.pdf.
- "Bethlehem Steel, The People Who Built America." Video, 1:19:27. YouTube. Posted by PBS39, February 5, 2008. Accessed January 29, 2024.
<https://www.youtube.com/watch?v=2QTGiHOZZFU>.
- Brodsky, Nathan. "The Weapons Acquisition Process — an Economic Analysis, by Merton J. Peck and Frederic M. Scherer. Boston: Division of Research, Harvard Business School, 1962. Pp. 736." *Naval Research Logistics Quarterly* 9, nos. 3-4 (1962): 619.
<https://doi.org/10.1002/nav.3800090313>.
- "Brownsville General Hospital." Abandoned Online. Accessed January 29, 2024.
<https://abandonedonline.net/location/brownsville-general-hospital/>.
- Brugeman, Valerie Sathe, Kim Hill, and Joshua Cregger. "Repurposing Former Automotive Manufacturing Sites." CAR (Center for Automotive Research). Last modified November 2011. Accessed November 13, 2023.
<https://www.cargroup.org/wp-content/uploads/2017/02/Repurposing-Former-Automotive-Manufacturing-Sites.pdf>.
- Center for Community Progress Report to the Urban Redevelopment Authority of Pittsburgh, Pennsylvania. "THE COST OF VACANT AND BLIGHTED PROPERTIES IN PITTSBURGH: A Conservative Analysis of Service, Tax Delinquency, and Spillover Costs." Center for Community Progress. Last modified March 2017. Accessed January 29, 2024.
<https://pghlandbank.org/wp-content/uploads/2022/03/Cost-of-Blight-2.pdf>.
- Christopher, Matthew. "Delaware River Generating Station." Abandoned America. Last modified September 24, 2021. Accessed January 29, 2024.
<https://www.abandonedamerica.us/delaware-river-generating-station>.
- . "Lincoln Way." Abandoned America. Last modified May 15, 2023. Accessed January 29, 2024. <https://www.abandonedamerica.us/lincoln-way>.
- CITY OF BETHLEHEM. "Comprehensive Plan 2008 Future Land Use." Bethlehem Pennsylvania. Last modified 2008. Accessed January 29, 2024.
https://www.bethlehem-pa.gov/CityOfBethlehem/media/DCED-Media-Library/PlanningZoning/CommunityPlans/Comprehensive_Plan_2009_Future_Land_Use.pdf.

- Collaton, Elizabeth, and Charles Bartsch. "Industrial Site Reuse and Urban Redevelopment—An Overview." *Cityscape* 2, no. 3 (1996): 17-61. <http://www.jstor.org/stable/20868420>.
- Culey, Sean. "Revitalizing the Rust Belt." *Forbes*. Last modified September 8, 2017. Accessed January 29, 2024. <https://www.forbes.com/sites/realspin/2017/09/08/revitalizing-the-rust-belt/>.
- Cusumano, Michael A. "Manufacturing Innovation: Lessons from the Japanese Auto Industry." *MIT Sloan Management Review*. Last modified October 15, 1988. Accessed January 29, 2024. <https://sloanreview.mit.edu/article/manufacturing-innovation-lessons-from-the-japanese-auto-industry/>.
- Department of City Planning. "Pathways to Removing Obstacles to Housing (PRO Housing)." The City of Pittsburgh. Accessed January 29, 2024. <https://engage.pittsburghpa.gov/pathways-removing-obstacles-housing>.
- Department of Environmental Conservation. Environmental Clean-up at Former Bethlehem Steel Site. <https://www.dec.ny.gov/chemical/122928.html>.
- Desmond, Mike. "Agreement expected this summer for \$69-\$200M Bethlehem Steel cleanup." *WBFO npr*. Last modified May 19, 2021. Accessed August 31, 2023. <https://www.wbfo.org/environment/2021-05-19/agreement-expected-this-summer-for-69-200m-bethlehem-steel-cleanup>.
- . "Agreement expected this summer for \$69-\$200M Bethlehem Steel cleanup." *WBFO*. Last modified May 19, 2021. Accessed January 29, 2024. <https://www.wbfo.org/environment/2021-05-19/agreement-expected-this-summer-for-69-200m-bethlehem-steel-cleanup>.
- Donaldson, Rachel, Ph.D. "Labor History in the United States." National Parks Service U.S. Department of the Interior. Last modified January 2022. Accessed January 29, 2024. https://portal.ct.gov/-/media/DECD/Historic-Preservation/03_Technical_Assistance_Research/Research/Labor_History_in_US-Theme_Study-Final-2.pdf.
- Downey, Kirstin. "Bethlehem Steel Corp. Is No More." *The Washington Post*. Last modified May 1, 2003. Accessed November 3, 2023. <https://www.washingtonpost.com/archive/business/2003/05/01/bethlehem-steel-corp-is-no-more/55b6daa3-9016-4ebd-9b27-b4b8c704afbc/>.
- Durbin, Dee-Ann. "Toyota passes Ford as 2nd-biggest auto seller in US in 2007." *LA Times*. Last modified January 3, 2008. Accessed January 29, 2024. <https://www.latimes.com/business/autos/la-ap-mi-autosales-story.html>.
- Fraser, Jeffery. "Opportunity Awaits but..." *Pittsburgh Today*. Last modified 2022. Accessed November 13, 2023. <https://pittsburghquarterly.com/articles/opportunity-awaits-but/>.

- Hahn, Ashley. "New chapter possible for Delaware Generating Station." WHYYPBS. Last modified January 20, 2015. Accessed January 29, 2024.
<https://whyy.org/articles/new-chapter-possible-for-delaware-generating-station/>.
- Harvard Business School. "Documenting the Wartime Effort." Baker Library Bloomberg Center.
<https://www.library.hbs.edu/us-steel/exhibition/documenting-the-wartime-effort>.
- Henshaw, Marc Nicholas. "The Steamboat Industry in Brownsville, Pennsylvania: An Ethnohistorical Perspective on the Economic Change in the Monongahela Valley." Master's thesis, Western Michigan U, 2004. Accessed August 31, 2023.
https://scholarworks.wmich.edu/masters_theses/3926/.
- IMF. External Relations Dept. "Finance & Development, December 1987." *Finance & Development* 24, no. 4 (1987). <https://doi.org/10.5089/9781616353711.022>.
- . "Finance & Development, December 1987." Abstract. *Finance & Development* 24, no. 4 (1987). <https://doi.org/10.5089/9781616353711.022>.
- Levinson, Marc. "Container Shipping and the Decline of New York, 1955-1975." *The Business History Review* 80, no. 1 (2006): 49-80.
<http://www.jstor.org/stable/25097151>.
- Licht, Walter. "Workshop of the World." The Encyclopedia of Greater Philadelphia. Accessed January 29, 2024.
<https://philadelphiaencyclopedia.org/themes/workshop-of-the-world/>.
- Manisalidis, Ioannis, Elisavet Stavropoulou, Agathangelos Stavropoulos, and Eugenia Bezirtzoglou. "Environmental and Health Impacts of Air Pollution: A Review." *Frontiers in Public Health* 8 (February 20, 2020).
<https://doi.org/10.3389/fpubh.2020.00014>.
- Marcus, Irwin M. "THE DEINDUSTRIALIZATION of AMERICA: HOMESTEAD, a CASE STUDY, 1959–1984." *Pennsylvania History: A Journal of Mid-Atlantic Studies* 52, no. 3 (1985): 162-82. <http://www.jstor.org/stable/27773053>.
- Mele, Chris. "Power plants on the Delaware: Abandoned but still celebrated." Delaware Currents. Last modified November 22, 2021. Accessed January 29, 2024.
<https://delawarecurrents.org/2021/11/22/power-plants-on-the-delaware-abandoned-but-still-celebrated/>.
- Metz, Lance E. "Bethlehem Steel: The Rise and Fall of an Industrial Giant." *Pennsylvania Legacies* 6, no. 2 (2006): 10-15. <http://www.jstor.org/stable/27765048>.
- Monsbrucker-Garza, Kristen. "Few are working full-time in the office anymore, but some argue it's nowhere near a 'real estate recession' in Philly." WHYYPBS. Last modified January 3, 2024. Accessed January 29, 2024.
<https://whyy.org/articles/center-city-office-market-real-estate-recession-2024-forecast/>.

- Moran, Tim. "VW production tripped up in US - Many problems lead to automaker's closing of PA factory." *Automotive News*. Last modified October 17, 2005. Accessed January 29, 2024.
<https://www.autonews.com/article/20051017/SUB/510170703/vw-production-tripped-up-in-u-s>.
- Munz, Aaron. "Bethlehem Steel Shipyard." Edited by Walter Pacheco. *Asbestos.com*. Last modified August 25, 2023. Accessed January 29, 2024.
<https://www.asbestos.com/shipyards/bethlehem-steel/>.
- November 1929*. *Popular Science*. Accessed January 29, 2024.
<https://books.google.com/books?id=FSgDAAAAMBAJ&q>.
- Parsons. "Frequently Asked Questions." Former Bethlehem Steel Site Cleanup. Accessed January 29, 2024. <https://bethlehemsteelcleanup.com/faq/>.
- "Philadelphia in the late 1800s: International Industrial Leader – Then and Now?" Historical Society of Pennsylvania. Last modified November 22, 2017. Accessed January 29, 2024.
<https://hsp.org/blogs/educators-blog/philadelphia-late-1800s-international-industrial-leader-%E2%80%93-then-and-now>.
- The Post Standard. "A timeline of Crucible Materials Corp." *Syracuse.com*. Last modified May 6, 2009. Accessed January 29, 2024.
https://blog.syracuse.com/indepth/2009/05/a_timeline_of_crucible_materia.html.
- RMC-Environmental Services. "Limerick Generating Station Units 1 & 2." License Renewal Company. Last modified December 1985. Accessed January 29, 2024.
<https://www.nrc.gov/docs/ML1211/ML12110A289.pdf>.
- "The Root of Pittsburgh's Population Drain." Carnegie Mellon Heinz School Center for Economic Development. Last modified November 2003. Accessed January 29, 2024.
<https://www.heinz.cmu.edu/ced/file/pop-drain.pdf>.
- Silvia, Stephen J. "ORGANIZING GERMAN AUTOMOBILE PLANTS IN THE USA - An Assessment of the United Auto Workers' Efforts to Organize German-Owned Automobile Plants." Hans Bockler Stiftung. Last modified December 2016. Accessed January 29, 2024. https://www.boeckler.de/fpdf/HBS-006530/p_study_hbs_349.pdf.
- Stradling, David, and Richard Stradling. "Perceptions of the Burning River: Deindustrialization and Cleveland's Cuyahoga River." *Environmental History* 13, no. 3 (2008): 515-35. <http://www.jstor.org/stable/25473265>.
- "Sunbelt Migration." University of Central Florida. Accessed January 29, 2024.
<https://vlp.cah.ucf.edu/popups/sunbelt.html>.
- Susser, I. "The Construction of Poverty and Homelessness in US Cities." *Annual Review of Anthropology* 25 (1996): 411-35. <http://www.jstor.org/stable/2155833>.

U.S. Census Bureau. "Decennial Census of Population and Housing." U.S. Census Bureau. Last modified December 11, 2013.

<https://www.census.gov/programs-surveys/decennial-census.html>.

US Government. "Summary of the Comprehensive Environmental Response, Compensation, and Liability Act (Superfund)." Environmental Protection Agency. Last modified September 6, 2023. Accessed January 29, 2024.

<https://www.epa.gov/laws-regulations/summary-comprehensive-environmental-response-compensation-and-liability-act>.

Weinmeyer, Richard, JD, MA, Ph.D. "The Safe Drinking Water Act of 1974 and Its Role in Providing Access to Safe Drinking Water in the United States." AMA Journal of Ethics. Last modified October 2017. Accessed January 29, 2024.

<https://journalofethics.ama-assn.org/article/safe-drinking-water-act-1974-and-its-role-providing-access-safe-drinking-water-united-states/2017-10>.

World Population Review.

<https://worldpopulationreview.com/us-cities/clairton-pa-population>.